



Day 1: Python Fundamentals

Name: Rohit Sanjay Bagal

Day: Sunday

Date: 09/08/2026

Cohort: AI-ML 2026

Contact: 9403780042

Email: rohitbagal1819@gmail.com

Description: Topics covered on Day 1

Topics Covered

1. Virtual Environment
 2. VSCode / Jupyter Notebook / Kaggle
 3. Commands
 4. Data Types
 5. Dictionary
 6. List
 7. Indexing
 8. Slicing
 9. Conditional Statements
 10. For Loop
 11. While Loop
-

Questions & Examples

Question 1 – Data Types

```
x = 1.2  
print(type(x))
```

Question 2 – User Input

```
x = int(input("Enter a valid number: "))  
print(x)
```

Question 3 – Type Conversion

```
a = "12345"  
b = int(a)  
print(b)
```

Question 4 – Dictionary

```
dict = {"x": "shraddha", "y": "rohit", "z": "mayank", "a": "rohit"}  
print(dict["x"])
```

Question 5 – List & Accessing Items

```
list = ["rohit", 2.5, 1000]  
print(list[1])
```

Question 6 – Slicing

```
a = "Here im learning new Technology"  
print(a[2:])
```

```
print(a[0:6])
print(a[0:13])
print(a[:13])
```

Question 7 – If-Else (Conditional Statements)

```
a = int(input("enter the age: "))
if a > 18:
    print("you can drive")
else:
    print("you cannot")
```

Question 8 – Advanced If-Elif-Else

```
a = int(input("Enter the total attendance of deboistech: "))
if a >= 80:
    print("You can get the certificate")
elif a >= 60:
    print("First complete your attendance")
else:
    print("you cannot getting a certificate")
```

Homework

For Loop

Example 1 – Print 1 to 10

```
for i in range(1, 11):
    print(i)
```

Example 2 – Iterate over a list

```
lists = ["ramu", "shamu", "rohu", "gauru", "sakshu"]
for item in lists:
    print(item)
```

Example 3 – Print even numbers from 1 to 20

```
for i in range(1, 21):
    if i % 2 == 0:
        print(i)
```

While Loop

Example 1 – Print 1 to 10

```
i = 1
while i <= 10:
    print(i)
    i += 1
```

Example 2 – Iterate over a list

```
students = ["rohit", "gaurav", "kunal", "vaibhav"]
i = 0
while i < len(students):
    print(students[i])
    i += 1
```

Additional Concepts

String ↔ Number Conversion

```
# String to int
a = "12345"
b = int(a)
```

```
print(b)

# Int to string
a = 12345
b = str(a)
print(type(b))
```

Tuple (Immutable)

A tuple is **immutable** — its elements cannot be changed after declaration. Represented with parentheses `()`.

```
subject = ("java", "python", 99, "dbms", "os")
print(subject[0])
print(subject[2])
print(subject[3])
```

List (Mutable)

A list is **mutable** — its elements can be changed. Represented with square brackets `[]`.

```
list = ["ramu", "shamu", True, 2.44, 5, "jayshivraj"]
print(list[0])
print(list[1])
print(list[2])
print(type(list[2]))
```